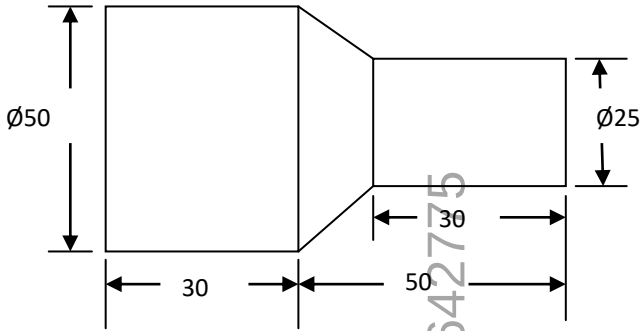


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE						
Supplementary Winter Examination – 2024						
Course: B.Tech.		Branch : Mechanical Engineering		Semester : VII		
Subject Code & Name: BTMEC702, CAD/CAM		/Mech. Engg. (Sandwich)				
Max Marks: 60		Date:07/02/2025		Duration: 3 Hr.		
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.						
					(Level /CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)					12
1	To view portion of a image enclosed in a rectangular region is called				CO-01	1
	a. View port	b. Windowing	c. Raster scan	d. Stroke writing		
2	To display different portions of the drawing in different region of the screen is called				CO-01	1
	a. View port	b. Windowing	c. Raster scan	d. Stroke writing		
3	In geometric modelling Solid model contains				CO-2	1
	a. Geometrical data	b. Topological information	c. Geometrical data & Topological information	d. Topographical information		
4	In geometric modeling wire frame and surface models contains				CO-2	1
	a. Geometrical data	b. Topological information	c. Geometrical data & Topological information	d. Topographical information		
5	Easiest and most advanced method of geometric modelling is				CO-2	1
	a. Solid modelling	b. Wire frame modelling	c. Surface modeling	d. None of these		
6	Primitives are combined by mathematical set of Boolean operations in				CO-03	1
	a. Constructive Solid modeling	b. Destructive Solid modeling	c. Boundary representation	d. Sweeping		
7	A surface model is generated by using wire frame entities - plane surface, ruled surface, tabulated surface & surface of revolution is known as				CO-03	1
	a. Analytical entities	b. Synthetic entities	c. Destructive Solid Geometry	d. Solid modelling		

8	Bezier surface allows				CO-03	1
	a. Local control	b. Global control	c. Both local and global control	d. None of these		
9	Elements of Numerical Control system are				CO-04	1
	a. Program of instructions	b. Machine control unit	c. Processing equipment	d. All the above		
10	In manual part program preparatory code for "rapid movement line path" is				CO-05	1
	a. G00	b. G01	c. G60	d. G71		
11	In manual part program code for auxiliary function "program end with rewind" is				CO-05	1
	a. M30	b. M01	c. M03	d. M02		
12	In APT programming the surface which guide the side of the cutter is known as				CO-06	1
	a. Drive surface	b. Part surface	c. Check surface	d. Fillet surface		
Q. 2	Solve the following.					12
A)	Explain minimum three cursor control devices used as CAD input device.				CO-01	6
B)	A triangle ABC with vertices A (20, 20), B (80, 20), and C (20, 80) is to be scaled by factor of 0.5 about a point X (40, 40). Determine composite transformation matrix and coordinates of vertices of scaled triangle.				CO-02	6
Q.3	Solve the following.					12
A)	Compare the CSG and B-Rep methods as a modeling technique for CAD.				CO-03	6
B)	What are the various windowing applications in CAD? Explain with the help of neat sketches and suitable examples.				CO-02	6
Q. 4	Solve Any Two of the following.					12

A)	Write a manual part program for a plain and taper turning the forged bar of 110 mm diameter as per the drawing shown. 	CO-05	6
B)	Classify numerical control system based on: 1. Type of control systems 2. Type of motion control 3. Number of axes Briefly discuss advantages and disadvantages of any one NC system.	CO-05	6
C)	What are the advantages of 'Point to point control' in CNC systems? Also mention in which particular applications it will be recommended?	CO-05	6
Q.5	Solve Any Two of the following.		12
A)	Explain the terms: A. Field variables B. Shape function C. Stiffness matrix	CO-06	6
B)	Explain various properties of stiffness matrix	CO-06	6
C)	Explain various advantages of Finite Element Methods	CO-06	6
Q. 6	Solve Any Two of the following.		12
A)	Discuss types of flexible manufacturing system layouts.	CO7	6
B)	What is computer integrated manufacturing (CIM)? List its benefits.	CO8	6
C)	Briefly explain Retrieval & Generative CAPP systems.	CO9	6
	*** End ***		